V-OP by PPEI

The Code Reader Is Dead. The Vehicle Optimizer Has Arrived.

This Is Not a Scan Tool.

Every scan tool on the market — from the $20Bluetoothdongleat AutoZonetothe$15,000 Snap-on ZEUS — does the same thing. It reads a five-character code. It displays a one-line description. It leaves you to figure out the rest.

V-OP does not read codes. V-OP reads the vehicle.

V-OP is the first AI-powered vehicle optimization platform built by the tuners who actually calibrate the software inside your ECU. Built by PPEI — the same team that has written more custom Duramax, Power Stroke, Cummins, and CAN-am calibrations than any independent tuner in the world. This is not a diagnostic tool designed by a software company that has never turned a wrench. This is a diagnostic tool designed by the people who write the calibration tables your ECU runs on.

Two Hardware Models. One Platform. Every Vehicle.

V-OP Lite

The smartest diagnostic tool ever built — for everyone.

V-OP Lite connects to any OBD-II vehicle through a standard ELM327 adapter or Bluetooth dongle. No installation. No subscription. No laptop required. It runs in your web browser on any phone, tablet, or computer.

What V-OP Lite does that no code reader can:

It monitors up to 20 live parameters simultaneously and runs real-time AI pattern analysis across every data point. It does not wait for the ECU to set a fault code. It watches the data and identifies developing problems before they become failures — before a code ever sets, before the check engine light ever illuminates, before the breakdown ever happens.

What a Code Reader Shows You	What V-OP Lite Shows You
P0299 — Turbo Underboost	VGT duty cycle commanding 92% but boost only reaching 24 PSI at 3,200 RPM — vane position not tracking command. Carbon buildup on unison ring. Clean or replace actuator.
P0087 — Fuel Rail Too Low	Rail pressure drops 1,200 PSI below command during 3-4 second WOT pulls above 3,000 RPM but recovers at cruise. CP4 pump volume insufficient under peak demand — not a sensor fault, not a regulator fault. Pump replacement required.
No code. Vehicle “runs fine.”	TCC slip averaging 38 RPM at 65 mph lockup with 14% duty cycle variation. Converter clutch material degrading. Transmission failure within 15,000 miles if not addressed.
P0401 — EGR Flow Insufficient	EGR valve commanding 45% open but position sensor reads 12%. Valve physically stuck — not an electrical fault. Intake manifold temps confirm reduced flow. Carbon cleaning required, not a new EGR valve.

That table is the difference between a $20codereaderanda$10,000 master technician’s brain. V-OP Lite puts that brain in your pocket for a fraction of the cost.

V-OP Lite Feature Set:

- **366 PIDs across 8 manufacturer platforms** — Standard OBD-II, GM Duramax extended, Ford Power Stroke extended, RAM Cummins extended, BMW UDS, CAN-am/Rotax, Sea-Doo/BRP Marine, Polaris, and Kawasaki. More parameters than a Snap-

on ZEUS reads, across more platforms, with zero annual subscription.

- **AI Diagnostic Engine** — 20+ pattern-based fault detection conditions with sustained deviation analysis, transient exclusion (ignores throttle tip-in spikes, gear shift transients, cold start anomalies), and multi-parameter cross-referencing that eliminates false positives. This engine was tuned against thousands of real-world datalogs until it achieved zero false positives on known-good vehicles while catching every real fault on known-bad vehicles.
- **Erika — Your AI Technician** — Not a chatbot. Not a search engine. Erika is a server-side AI with a curated knowledge base that includes the complete PID database, UDS reference library, module address maps for 80+ ECU modules, CAN bus protocol expertise, vehicle coding procedures, service procedure knowledge, and the full NRC code reference. Ask her anything about your vehicle and she answers with the specificity of a master technician who has memorized every service manual, every TSB, and every calibration table ever written.
- **Highlighted Fault Graphs** — When V-OP detects a fault condition, it does not just flag it with a red icon. It highlights the exact region on the data graph where the fault occurred, overlays the threshold boundaries, and draws attention to the specific parameter relationships that triggered the detection. You see the fault in context — the RPM, the load, the temperature, the duration — not just a code number.
- **AI Reasoning Engine** — Every diagnostic finding comes with a full reasoning chain. V-OP does not just say "rail pressure fault detected." It shows you the logic: "Rail pressure command was 26,500 PSI. Actual rail pressure was 25,300 PSI. Delta of 1,200 PSI sustained for 3.2 seconds during WOT above 3,000 RPM. This pattern is consistent with CP4 pump volume loss under peak demand. The fault did not occur at idle or cruise, ruling out regulator and sensor faults. Confidence: 94%." Every diagnosis is transparent. Every conclusion is justified. No black boxes.
- **Dyno Charts** — Upload any datalog and V-OP generates estimated horsepower and torque curves from OBD-II data. Not a replacement for a chassis dyno, but a real-time performance baseline that lets you see the effect of every modification, every tune revision, and every maintenance item. Overlay multiple runs. Compare before and after. See exactly where power is being made and where it falls off.
- **Drag Strip Timeslips** — V-OP analyzes acceleration data from your datalogs and generates virtual quarter-mile timeslips with 60-foot, 330-foot, 660-foot (eighth-mile), 1000-foot, and quarter-mile times with trap speeds. Compare runs. Track improvements. Know your vehicle's performance without paying for a track day.
- **Vehicle Health Reports** — One-click PDF generation with a scored health assessment, parameter-by-parameter analysis, fault findings with severity ratings, and actionable recommendations. Hand this to a customer and they understand exactly what their vehicle needs. Hand this to a shop and they have a diagnostic roadmap. Professional, branded, and comprehensive.
- **Multi-Vehicle Support** — V-OP auto-detects the vehicle platform via VIN decode and loads the correct PID set, diagnostic thresholds, and manufacturer-specific knowledge. Duramax one day, Power Stroke the next, CAN-am on the weekend. One tool. Every vehicle.
- **Powersports and Marine** — CAN-am, Sea-Doo, Polaris, and Kawasaki support with manufacturer-specific PIDs that no other consumer or professional diagnostic tool on the market can access. Rotax ECU temperatures, turbo boost, DESS key status, CVT belt temps, supercharger boost, impeller RPM, AWD engagement, gear position — all live, all analyzed, all explained by Erika.

V-OP Pro

The tool that replaces the dealer. And the tuner's laptop. And the $15,000 scan tool.

V-OP Pro includes everything in V-OP Lite plus professional-grade vehicle programming, ECU calibration flashing, ECU unlocking, and full UDS diagnostic services. It connects through a PEAK PCAN-USB device for direct CAN bus access with raw frame-level control.

V-OP Pro adds:

- **Calibration Flashing** — Read and write ECU calibration files directly through the OBD port. Flash PPEI custom tunes, revert to stock calibrations, update calibration revisions — all from the V-OP interface. No separate flashing software. No separate

licensing. The same platform that diagnoses your vehicle also tunes it. This is what HP Tuners, EFILive, and bFlash do — but integrated into a complete diagnostic and optimization platform instead of a standalone flash tool.

- **ECU Unlocking** — Locked ECU? V-OP Pro includes security access procedures for supported platforms. GM CMAC authentication, Ford LFSR seed/key, Cummins byte-swap/rotate/XOR, CAN-am matrix lookup, Polaris polynomial — the algorithms are built in. Where manufacturer security allows, V-OP Pro handles the unlock sequence automatically. Where newer security (like the post-2022.5 MG1CA920 HSM lockdown on CAN-am) blocks programmatic access, V-OP Pro tells you exactly what the ECU rejected and why, so you know the precise security boundary you are hitting.
- **Vehicle Coding** — Read and write ECU configuration data at the bit level. Ford IPC as-built blocks for fuel tank size changes (23 gallon to 80 gallon — every size, including aftermarket auxiliary tanks), tire size correction (speedometer recalibration for any tire size with exact revolutions-per-mile calculation), DRL configuration, auto-lock behavior, and more. RAM BCM configuration for fuel tank and tire size. This is what FORScan does — but with a guided UI instead of raw hex editing, and with AI assistance that explains what every bit does before you change it.
- **CAN-am VIN Programming** — Full VIN change procedure for CAN-am vehicles using the PEAK device. Step-by-step wizard: connect to ECU, read current VIN, enter extended diagnostic session, perform security access with seed/key challenge-response, write new VIN via UDS WriteDataByIdentifier, reset ECU, verify. Supports both MED17.8.5 and MG1CA920 ECU variants with automatic detection. Includes DESS key re-learn guidance.
- **Service Procedures** — Seven guided UDS service procedures that currently require a dealer scan tool or specialized software:

Procedure	What It Does	Current Cost at Dealer
DPF Forced Regeneration	Commands a stationary desoot cycle to burn off accumulated soot	$150-300$
Injector Coding	Writes IQA/IMA trim codes after injector replacement	$100-200$ per injector
TPMS Sensor Relearn	Programs new sensor IDs into the BCM after tire/sensor replacement	$50-100$
Transmission Adaptive Reset	Clears learned shift points and fuel trims for fresh adaptation	$75-150$
Oil Life Reset	Resets the oil life monitor after service	$0-50$ (often bundled)
Throttle Body Alignment	Relearns throttle body position after cleaning or replacement	$50-100$
Steering Angle Calibration	Calibrates the steering angle sensor after alignment or suspension work	$75-150$

Every one of these procedures follows the same UDS protocol flow: enter extended diagnostic session ($1003), performsecurityaccessifrequired(27)$, execute the service ($2Eor31$), reset the ECU ($11), verify. V-OP Pro walks you through each step with real-time feedback, NRC error translation, and Erika standing by to explain anything that goes wrong.

- **IntelliSpy — AI-Powered CAN Bus Sniffer** — Real-time CAN bus frame capture with automatic module identification, byte-level change highlighting, per-ID frequency analysis, and three analysis views: live scrolling trace, statistical breakdown, and AI-assisted decode. IntelliSpy cross-references every arbitration ID against the 80-module database, identifies which ECU sent each frame, and when unknown IDs appear, analyzes the data pattern to suggest what the signal likely represents. This is Vehicle Spy and SavvyCAN combined — with an AI brain that helps you reverse-engineer proprietary protocols in real time.
- **Module Scanner** — Scan every ECU address on the CAN bus (0x700-0x7FF), read identification DIDs (part number, hardware version, software version, VIN, serial number), and build a complete module map of the vehicle. Know exactly what ECUs are present, what software they are running, and whether any modules are offline or not responding. This is the first step in any professional diagnostic workflow, and V-OP Pro does it in seconds.

The Competitive Landscape Is Not Even Close

Capability	Code Reader	Snap-on ZEUS	FORScan	HP Tuners	V-OP Lite	V-OP Pro
Price	20–200	$10,000+	50–100	500–800	**TBD**	**TBD**
Annual Subscription	None	1,000–3,000/yr	None	200–500/yr (credits)	**None**	**None**
Read DTCs	Yes	Yes	Yes	Yes	**Yes**	**Yes**
Clear DTCs	Some	Yes	Yes	Yes	**Yes**	**Yes**
Live Data (Standard PIDs)	Limited	Yes	Yes	Yes	**Yes (96 PIDs)**	**Yes (96 PIDs)**
Manufacturer Extended PIDs	No	Partial	Ford/FCA only	GM only	**Yes (366 PIDs, 8 platforms)**	**Yes (366 PIDs, 8 platforms)**
AI Fault Analysis	No	No	No	No	**Yes (20+ conditions)**	**Yes (20+ conditions)**
AI Reasoning Engine	No	No	No	No	**Yes (transparent logic chains)**	**Yes (transparent logic chains)**
Highlighted Fault Graphs	No	No	No	No	**Yes**	**Yes**
AI Assistant (Erika)	No	No	No	No	**Yes**	**Yes**
Dyno Charts	No	No	No	Yes (with dyno)	**Yes (OBD-II estimated)**	**Yes (OBD-II estimated)**
Drag Timeslips	No	No	No	No	**Yes**	**Yes**
Vehicle Coding (Tank/Tire)	No	Dealer only	Yes (raw hex)	No	No	**Yes (guided UI)**
Calibration Flashing	No	No	No	Yes	No	**Yes**
ECU Unlocking	No	No	No	Partial	No	**Yes (multi-platform)**
VIN Programming	No	Dealer only	No	No	No	**Yes (CAN-am)**
Service Procedures	No	Yes (subscription)	Partial	No	No	**Yes (7 procedures)**
CAN Bus Sniffing	No	No	No	No	No	**Yes (IntelliSpy)**
Module Scanner	No	Yes	Partial	No	No	**Yes (80+ modules)**
Powersports Support	No	No	No	No	**Yes (4 platforms)**	**Yes (4 platforms)**
Health Report / PDF	No	Basic	No	Basic	**Yes (scored, detailed)**	**Yes (scored, detailed)**
Runs in Browser	No	No	No	No	**Yes**	**Yes**
No Installation Required	Some	No	No	No	**Yes**	**Yes**

Read that table again. There is no column where any existing tool wins. Not one. V-OP Lite alone outperforms every code reader and most professional scanners. V-OP Pro outperforms everything on the market combined — including tools that cost 10 to 50 times more.

Who Needs V-OP

Walmart Auto Centers — 2,500+ Locations, 75,000+ Technicians

A customer rolls in for a tire rotation. The check engine light is on. Today, the Walmart technician plugs in a $50 code reader, writes "P0420 — Catalyst Efficiency Below Threshold" on a sticky note, and tells the customer to take it to a mechanic.

With V-OP Lite, that same technician runs a 5-minute vehicle health scan while the tires are being rotated. V-OP analyzes the O2 sensor data, identifies that the downstream O2 sensor is switching too slowly (not a failed catalytic converter — a $2,000 repair — but a lazy O2 sensor — a$150 repair), generates a branded PDF health report, and hands it to the customer. The customer gets the right diagnosis. Walmart gets a reputation for actually helping people. The upsell opportunity is enormous.

Truck Stops — Every Minute of Downtime Costs $500-$1,000

A Class 8 diesel pulls into a Pilot/Flying J with a DPF warning light. The truck stop mechanic plugs in a code reader. "P2463 — DPF Soot Accumulation." Does the truck need a $200 forced regen or a$5,000 DPF replacement? The code reader cannot tell you. The mechanic guesses. If he guesses wrong, the driver is stuck for two days waiting for parts.

V-OP Lite analyzes the DPF soot load percentage, exhaust gas temperatures, regen history, DEF quality, and backpressure data. It determines that soot load is at 85% but EGTs are reaching regen temperature normally — the truck just needs a forced regen, not a new DPF. With V-OP Pro, the mechanic commands the forced regen right there in the service bay. Driver is back on the road in 30 minutes. Trucking company saves thousands.

Tire Shops — A $100 Upsell on Every Tire Size Change

Customer installs 35-inch tires on their F-250 Power Stroke. Speedometer now reads 10% slow. Currently, the customer has to go to the dealer ($150+$), buy a FORScan license and OBD cable and figure out the editing (100+ and hours of forum reading), or just live with an inaccurate speedometer (which also throws off the odometer, fuel economy calculations, and transmission shift points).

V-OP Pro reads the current IPC as-built data, shows the customer exactly what their speedometer is calibrated for, calculates the correction factor for their new tire size, and writes the corrected value — all through a guided UI that takes 5 minutes. The tire shop charges $50-$100 for the service. Pure profit. Every single tire size change is an upsell opportunity.

Independent Mechanic Shops — 160,000+ in the United States

The average independent shop spends $5,000-$15,000 on scan tools plus $1,000-$3,000 per year in subscription renewals. They get a tool that reads codes, shows some live data, and occasionally runs a guided test. The technician still has to know what the data means.

V-OP Lite provides equivalent or superior diagnostic capability with AI analysis that explains what the data means. V-OP Pro adds the programming and service procedure capabilities that currently require a dealer scan tool. One tool replaces the Snap-on, the Autel, the FORScan laptop, and the HP Tuners rig. One tool. No subscription. No annual renewal. No per-vehicle licensing.

Everyday Vehicle Owners — 290 Million Registered Vehicles

The check engine light comes on. The owner has three options: ignore it and hope it goes away, pay a shop $100-$150 for a diagnostic, or buy a $20 code reader and Google the code. None of these options are good.

V-OP Lite connects through a $15 Bluetooth ELM327 adapter from Amazon. The owner opens the web app on their phone. V-OP reads the vehicle, analyzes the data, identifies the problem, explains it in plain language through Erika, and generates a health r
150. They have the data. They have the analysis. They have the power.

Diesel Performance Shops — The Core Market

This is where V-OP was born. PPEI tunes more diesels than anyone. The shops that install those tunes need a diagnostic tool that understands tuned vehicles — because every other scan tool on the market flags tuned parameters as faults. A tuned L5P running 40 PSI of boost is not "overboost." A tuned Cummins with 30,000 PSI rail pressure is not "rail pressure too high." V-OP understands the difference between a fault and a tune because it was built by the people who write the tunes.

V-OP Pro is the complete shop tool: diagnose the customer's truck, flash the PPEI tune, verify the tune is running correctly, run a virtual dyno pull to confirm power output, generate a before/after comparison report, and hand the customer a professional PDF showing exactly what changed. No other tool in the world does all of that in one platform.

The Technology Stack — Built Different

V-OP is not a mobile app with a Bluetooth connection and a DTC database. It is a full-stack AI platform with seven interconnected intelligence layers:

Layer 1 — Data Acquisition. Multi-protocol connection engine supporting ELM327 (Bluetooth/WiFi/USB), PEAK PCAN-USB (direct CAN bus), and WebSerial. Configurable sample rates. Multi-PID concurrent polling. Raw CAN frame access for IntelliSpy and UDS operations.

Layer 2 — Signal Processing. Real-time data normalization, unit conversion, derived parameter calculation (estimated HP/torque from MAF and timing data), and downsampling for visualization. Handles manufacturer-specific scaling factors and encoding formats automatically.

Layer 3 — Pattern Analysis. 20+ diagnostic fault conditions with sustained deviation detection, transient exclusion, and multi-parameter cross-referencing. Each condition defines the specific parameter relationships, threshold values, minimum sample counts, and exclusion criteria that distinguish a real fault from normal operation.

Layer 4 — AI Reasoning. Server-side LLM integration that receives the full diagnostic context — vehicle identification, parameter data, detected fault patterns, calibration data if available — and generates human-readable explanations with transparent reasoning chains. Every conclusion cites the specific data points that support it.

Layer 5 — Vehicle Programming. Full UDS (ISO 14229) implementation: Diagnostic Session Control $(10), SecurityAccess(27)$ with multi-platform seed/key algorithms, Read/Write Data By Identifier $(22/2E)$, Routine Control $(31), IOControl(2F)$, ECU Reset ($11). This is the same protocol layer that dealer scan tools and professional flashing tools use.

Layer 6 — Knowledge Base. Curated automotive knowledge including PID definitions, DTC code database, module address maps, as-built data decode tables, service procedure sequences, NRC code reference, and CAN bus tool expertise. This is not scraped from forums. It is compiled from service manuals, calibration documentation, and decades of tuning experience.

Layer 7 — Visualization. Interactive charts with zoom, pan, and overlay. Highlighted fault regions with threshold boundaries. Dyno curves with HP and torque. Drag timeslips with split times. Health reports with scored assessments. Everything exportable to PDF with PPEI branding.

The Numbers

Metric	Value
Total PIDs	366 across 8 manufacturer platforms
Supported Vehicles	All OBD-II (1996+), CAN-am, Sea-Doo, Polaris, Kawasaki
ECU Modules Mapped	80+ (Ford 48, RAM 21, GM 11+)
Diagnostic Fault Conditions	20+ pattern-based detections
UDS Services Supported	8 (full ISO 14229 stack)
Seed/Key Algorithms	5 platforms (GM, Ford, Cummins, CAN-am, Polaris)
Service Procedures	7 guided procedures
Common Fuel Tank Sizes	23 (14 Ford + 9 RAM)
Common Tire Sizes	25 with revs/mile lookup
Automated Test Coverage	534 passing tests across 20 test suites
Hardware — V-OP Lite	Any ELM327 adapter (15–30)
Hardware — V-OP Pro	PEAK PCAN-USB ($250) + ELM327
Platform	Web browser — Chrome, Edge, Firefox, Safari
Installation Required	None
Annual Subscription	None

The Bottom Line

The automotive diagnostics industry has been selling the same product for 30 years: a device that reads a code and displays a description. The technology inside vehicles has advanced by orders of magnitude. The diagnostic tools have not.

V-OP by PPEI is the first tool that matches the intelligence of the vehicle it is diagnosing. It does not just read codes — it reads the vehicle. It does not just display data — it analyzes it, reasons about it, and explains it. It does not just identify problems — it identifies problems before they set codes, before the check engine light comes on, before the breakdown happens. And with V-OP Pro, it does not just diagnose — it programs, calibrates, flashes, unlocks, and optimizes.

V-OP Lite puts a master technician's diagnostic brain in the hands of every Walmart auto center, every truck stop, every tire shop, every independent mechanic, and every vehicle owner in the country. No training required. No subscription required. No technical knowledge required.

V-OP Pro puts a dealer scan tool, a professional flash tool, a CAN bus analyzer, and an AI diagnostic engine into a single platform that costs less than the annual subscription renewal on a Snap-on scanner. It is the only tool on the market that can diagnose a vehicle, flash a tune, verify the calibration, run a virtual dyno pull, and generate a professional report — all without leaving the browser.

The code reader is dead.

V-OP by PPEI. Redefining the Limits.

V-OP by PPEI — Vehicle Optimizer Built by tuners. Powered by AI. Made for everyone.

References